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Inherently Transparent Models

Global Explainability

Lecture: Explainable Artificial Intelligence, ST 2026

Lecturer: Jun-Prof. Gesina Schwalbe

Institute of Artificial Intelligence, Ulm University

Version: f5ee5d8, 2026-04-22 04:37:27 +0200

Outline

k-Nearest Neighbors

Linear Regression and Classification

- Linear Regression

- Explaining Linear Models: Reading out Feature Importance

- Reducing Features via LASSO Regression

- Logistic Regression for Classification

Decision Trees

- Decision Trees on Categorical Data

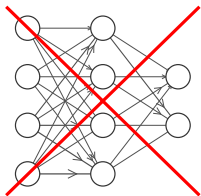
- Explaining Decision Trees

Evaluating the Quality of Transparent Models

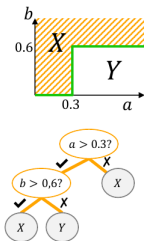
Summary

Revision

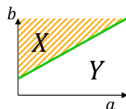
Fully *transparent* (simulatable) = understandable as a whole



› Decision **Trees**



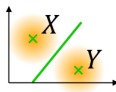
› **Linear** Models,
Generalized **Additive** Models



$$\text{Linear: } f(x) = \alpha a + \beta b$$

$$\text{GAM: } f(x) = g^{-1}(f_a(a) + f_b(b))$$

› **Clustering**



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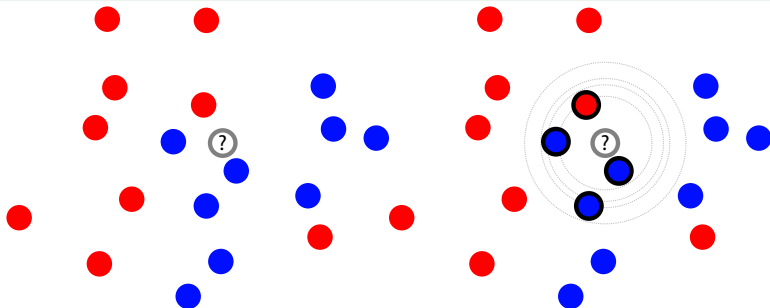
Summary

Instance-based: k-nearest neighbors (kNN)

Example (Assign colors to 2D points)

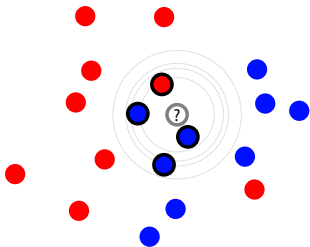
- Given:**
- ▶ (many) labeled data points $(x, y) \in D_{\text{train}} \subset X \times Y$,
e.g., $X = \mathbb{R} \times \mathbb{R}$, $Y = \{\text{red}, \text{blue}\}$.
 - ▶ distance metric $d: X \times X \rightarrow \mathbb{R}$, e.g., $d(a, b) := \|a - b\|_2$
 - ▶ new unlabeled point $x \in X$

Goal: inference for x , i.e., y matching x (here: color of x)



Pros and Cons of Instance-based Modelling

- + lazy
- + Easy updates (adding and *forgetting* of samples!)
- + explanation of reasoning easy (show examples)
- + decisions back-traceable to (relevance of) training data points
- Results strongly depend on k and metric
- High memory consumption
- $O(n)$ complex neighbor search



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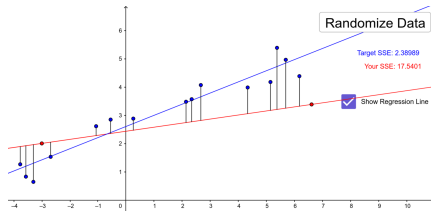
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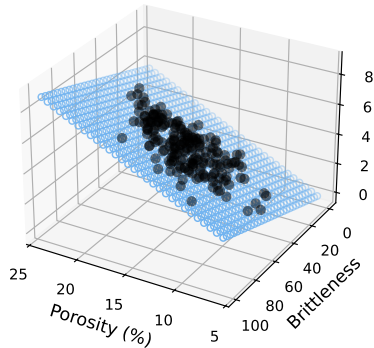
Summary

Linear Regression



- ▶ linear function $f: \mathbb{R}^n \rightarrow \mathbb{R}^1, f(x) = x^T \theta + \theta_0$
- ▶ $\theta \in \mathbb{R}^n, \theta_0 \in \mathbb{R}$ to be estimated from N samples
 $\mathbf{d} = ((x^{(i)}, y^{(i)}))_{i \leq N} \in \mathbb{R}^{N \times (n+1)}$
- ▶ s.t. $\text{SSE}(f; \mathbf{d}) = \sum_{(x,y) \in \mathbf{d}} \|y - f(x)\|^2$ minimal

Daily Gas Production [1000Mcf/day]



Solving for a Linear Regressor

has a *unique* solution!

$$\begin{aligned}\text{To minimize } \text{SSE}(f; \mathbf{d}) &= \sum_{(x,y) \in \mathbf{d}} \|x^T \theta - y\|^2 \\ &= \sum_{(x,y) \in \mathbf{d}} (x^T \theta - y)^T (x^T \theta - y) \\ &= (X\theta - Y)^T (X\theta - Y) = \theta^T X^T X \theta - Y^T X \theta - \theta^T X^T Y + Y^T Y \\ &= \theta^T \underbrace{X^T X}_{\mathbf{C} \in \mathbb{R}^{n \times n} :=} \theta - 2 \underbrace{Y^T X}_{=: -d \in \mathbb{R}^n} \theta + Y^T Y \quad (\text{convex in } \theta!)\end{aligned}$$

$$\text{calculate } \frac{\partial}{\partial \theta} \text{SSE} = 2 \underbrace{X^T X}_{\mathbf{C} \in \mathbb{R}^{n \times n} :=} \theta - 2 \underbrace{Y^T X}_{=: -d \in \mathbb{R}^n} \theta \stackrel{!}{=} 0$$

$\Rightarrow$ solve the linear equation system $\mathbf{C}\theta + d = 0$ for optimal θ^* .

An Example

Example

To fit $f_\theta: \mathbb{R}^1 \rightarrow \mathbb{R}^1$ we have $N = 4$ data points $\mathbf{d}$.

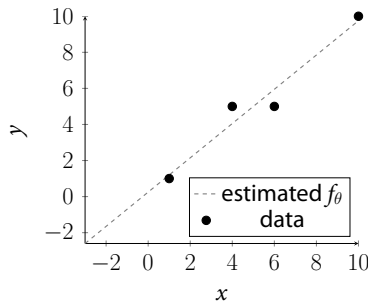
1. $X \leftarrow \begin{pmatrix} 1 & 1 \\ 1 & 4 \\ 1 & 6 \\ 1 & 10 \end{pmatrix} \in \mathbb{R}^{4 \times (1+1)}, Y \leftarrow \begin{pmatrix} 1 \\ 5 \\ 5 \\ 10 \end{pmatrix} \in \mathbb{R}^{4 \times 1}.$

2. $C = X^T X = \begin{pmatrix} 4 & 21 \\ 21 & 153 \end{pmatrix},$

$$d = -X^T Y = \begin{pmatrix} -21 \\ -151 \end{pmatrix}$$

3. $\theta \approx \begin{pmatrix} 0.25 \\ 0.95 \end{pmatrix}$ is solution to $C\theta + d = 0$
 $\rightarrow f_\theta(x) \approx 0.95x + 0.25.$

X	Y
1	1
4	5
6	5
10	10



What can be explained?

How: knowledge and “reasoning”:

- How certain is the choice of each slope θ_k ?

standard error: $SE(\theta_k) = \sqrt{\frac{N}{N-n} \frac{SSE(f_\theta; \mathbf{d})}{\text{Var}[X_k]}}$

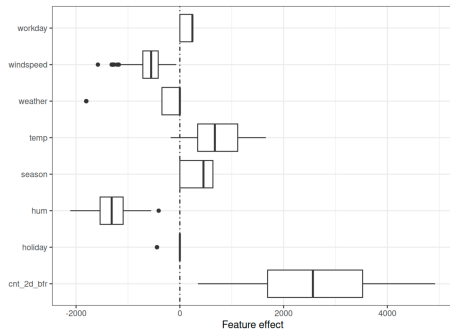
- What features are generally relevant?

Significance level under t-statistic: $t_{\theta_k} = \frac{\theta_k}{SE(\theta_k)}$

Why: What features were relevant for this decision?

Feature effects: $\text{effect}_k^{(i)} = \theta_k x_k^{(i)}$

weights alone are sensitive to feature scaling!



Regularization for Linear Models

Theorem

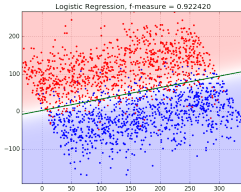
Given a deterministic function g_θ where $P(Y | X) = \mathcal{N}(g_\theta(X), \sigma)$
 $\theta_{MAP} = \operatorname{argmax}_\theta P(\theta | \mathbf{d}) = \operatorname{argmin}_\theta SSE(g_\theta; \mathbf{d}) - \lambda \log P(\theta).$

Remark (Regularization in Practice)

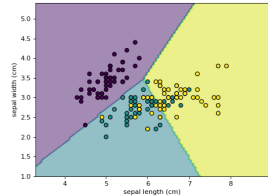
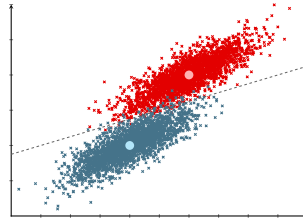
	Prior $P(\theta)$	Formula for θ_{MAP}
No ass.	uniform	$\operatorname{argmin}_\theta \sum_{(x,y) \in \mathbf{d}} \ x^T \theta - y\ _2^2 = \theta_{MLE}$
Ridge regression	normal	$\operatorname{argmin}_\theta \sum_{(x,y) \in \mathbf{d}} \ x^T \theta - y\ _2^2 + \lambda \ \theta\ _2^2$
Lasso regression	Laplace	$\operatorname{argmin}_\theta \sum_{(x,y) \in \mathbf{d}} \ x^T \theta - y\ _2^2 + \lambda \ \theta\ _1 \quad \Leftarrow \text{most para.s (near) zero}$

Logistic Regression for Classification

Linear regression



Linear classification



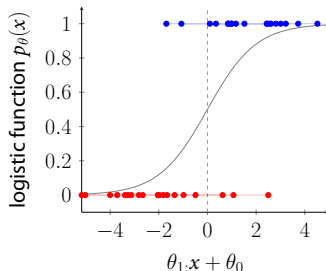
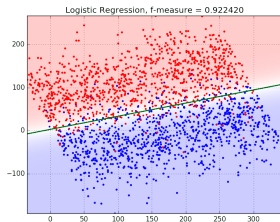
Examples: spam filter; hate speech detection; fraud detection; disease probability; ...

Logistic Regression for Classification

- **Goal:** For class c find a **linear hyperplane** with normal vector θ and bias θ_0 such that $f_\theta(x) = c \Leftrightarrow d_\theta(x) = \theta_1 x + \theta_0 > 0$.
- **Idea:** Make y continuous ($c = 1, \neg c = 0$) and fit *logistic* function

$$P(Y = c \mid x, \theta) = p_\theta(x) = \frac{\exp(d_\theta(x))}{1 + \exp(d_\theta(x))}$$

The *logit* d_θ linearly predicts distance from the hyperplane.



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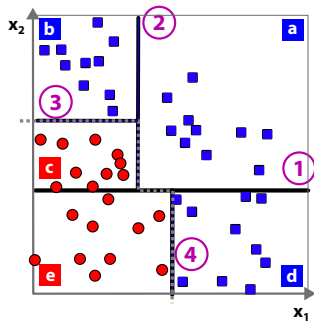
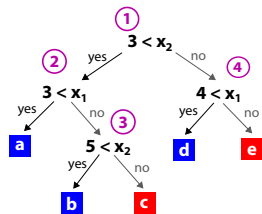
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Summary

Idea

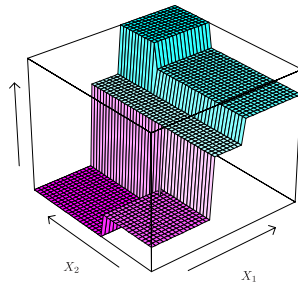
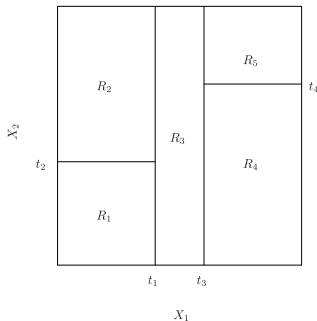
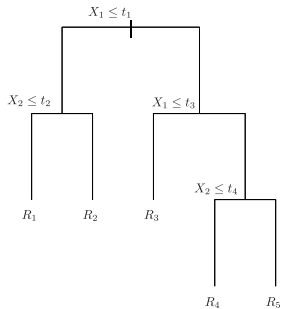
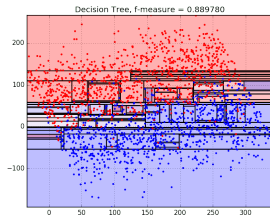


Good solar panel output? (x_1 = sun angle, x_2 = temperature)

- ▶ Idea: A node should **discriminate** its inflowing inputs **as much as possible**.
- ▶ General steps for **Top-down tree construction**:
 - ▶ **Partition** training samples **recursively**:
 1. choose best split **threshold(s)** per feature;
 2. choose best splitting **feature**; split.

Typical stopping criterion: max. depth, or # samples in leaf small
 - ▶ highlight Per leaf split **assign output**,
e.g., average / majority vote

The Shape of the Decision Surface

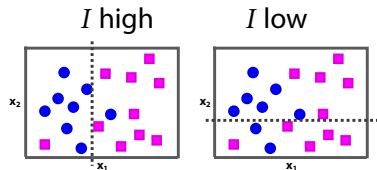


A Splitting Criterion

Idea: Compare entropy **before the split** to **after the split**.

Definition (Conditional Entropy)

$$H(Y | X) := - \sum_{\substack{y \in \text{values of } Y \\ x \in \text{values of } X}} P(x) P(y | x) \log_2(P(y | x))$$



Definition (Information Gain, Mutual Information)

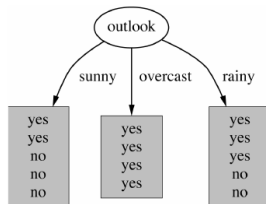
$$\begin{aligned} \text{(information gain)} \quad & \text{IG}(Y; x) := \overbrace{H(Y)}^{\text{entropy before}} - \overbrace{H(Y | X = x)}^{\text{entropy after (single split set)}} \\ \text{(mutual information)} \quad & I(X; Y) := E_{x \sim P(X)} [IG(Y; x)] = H(Y) - \underbrace{H(Y | X)}_{\text{entropy after (all split sets avg.ed)}} \end{aligned}$$

measures how much information (on average) X encodes about Y .

Example on Categorical data

Example

$$\begin{aligned} I(\text{Play}; \text{Outlook}) &= H(\text{Play}) - \sum_{o \in \{\text{sunny}, \text{overcast}, \text{rainy}\}} P(o) H(\text{Play} \mid o) \\ &= H(\text{Play}) \\ &\quad - P(\text{sunny}) H(\text{Play} \mid \text{sunny}) \\ &\quad - P(\text{overcast}) H(\text{Play} \mid \text{overcast}) \\ &\quad - P(\text{rainy}) H(\text{Play} \mid \text{rainy}) \\ &\approx 0.94 - \frac{5}{14} \cdot 0.97 - \frac{4}{14} \cdot 0 - \frac{5}{14} \cdot 0.97 \\ &\approx 0.25 \end{aligned}$$



Bonus: Other Splitting Criteria

Definition

For $S \subset \mathbf{d}$ the current split set:

Gini impurity $S|_{X=x}$ is less impure **the better guessing works:**

$$\text{Gini}(X = x; Y) := \sum_{y \in \text{values of } Y} \overbrace{P(y | x)}^{\text{prob. of guess being correct}} \overbrace{P(Y \neq y | x)}^{\text{prob. of guess being wrong}} = 1 - \sum_y P(y | x)^2$$

Variance reduction Choose X that *most reduces variance* on average:

$$\text{VarRed}(X; Y) := \overbrace{\frac{1}{|S|^2} \sum_{s, s' \in S} k(y_s, y_{s'})}^{\approx \text{Var}[Y]} - \sum_{x \in \text{vals of } X} \overbrace{\frac{1}{|S|^2} \sum_{s, s' \in S|_{X=x}} k(y_s, y_{s'})}^{\approx P(x) \text{Var}_{y \sim P(Y | x)}[Y]}$$

for a dissimilarity measure k , typically $k(y, y') = \frac{1}{2}(y - y')^2$.

applicable to **continuous output variables!**

Common Algorithms

1960s ID3 ^[5] (Quinlan)

1980s CART (Classification And Regression Tree) ^[3]
uses Gini impurity

1970s CHAID (Chi-squared Automatic Interaction Detector) ^[2]

1990s C4.5 ^[4] (Quinlan):

- ▶ Permits numeric attributes
- ▶ Deals with missing values
- ▶ Pruning to deal with noisy data

one of best-known and most widely-used learning algorithms;
continuously improved (commercial successor: C5.0)

Today implementations available in most frameworks, e.g., scikit-learn, R.

What can be explained?

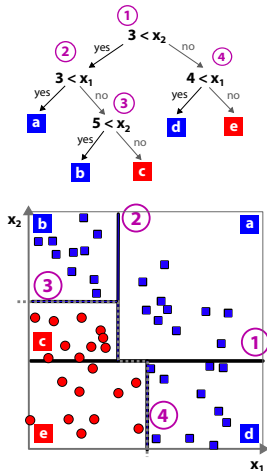
How: knowledge and “reasoning”:

- ▶ *What reasoning is learned & applied?*
tree structure
- ▶ *Which features were generally relevant?*
average local feature relevancies

Why: Traversing the tree:

- ▶ *What features were relevant for this decision?*
Aggregate split criterion improvements at each of the feature’s decision points.
- ▶ *What split decisions were relevant?*

$$f(x) = \underbrace{\bar{y}}_{\text{root pred.}} + \sum_{S \in \text{Splits}} \underbrace{\bar{y}_{S} - \bar{y}_{\text{Parent}(S)}}_{\text{contrib. of that split}}$$



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Goodness of Fit

For labeled test dataset $\mathbf{d}_{\text{test}} \subset X \times Y$, $Y \subset \mathbb{R}^n$ (numerical) or $Y \subset \{c_1, \dots, c_k\}$ (categorical), and function f :

Example (Regression Metrics)

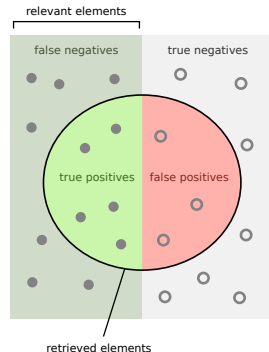
squared error: $SSE(f; D) := \sum_{(x,y) \in D} (y - f(x))^2$ $MSE(f; D) := \frac{1}{|D|} SSE(f; D)$

root mean squared error: $RMSE(f; D) = \sqrt{MSE(f; D)}$

Example (Classification Metrics)

$$\text{Accuracy}(f; D) = \frac{|\text{Correct Predictions}|}{|\text{All Predictions}|} = \frac{TP + TN}{(TP + FN) + (TN + FP)}$$

$$F_1 \text{ score}(f; D) = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 TP}{2 TP + FP + FN}$$



How many retrieved items are relevant?



How many relevant items are retrieved?



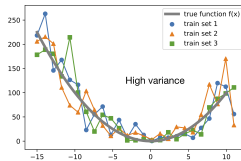
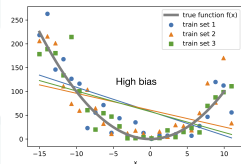
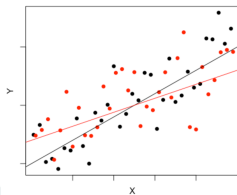
Bias-Variance Tradeoff

- What is the **average outcome** of $f(x)$? $\overline{f(x)} := E_{\mathbf{d} \sim P(\mathbf{D}^{(N)})} [f(x)]$
 $\Rightarrow$ How close is the average output $\overline{f(x)}$ to the optimal $\overline{y}(x)$?
Can my model type **approximate the optimal** solution?

$$\text{Bias } [f_{\hat{\theta}}] (x) := (\overline{y}(x) - \overline{f(x)})$$

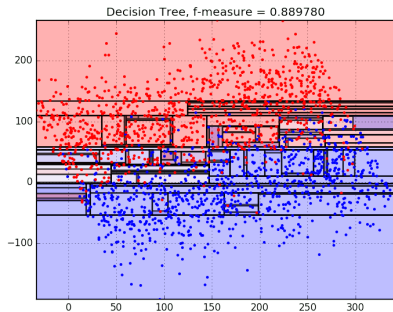
- How much do learned solutions f **vary** for different $\mathbf{d}$?

$$\text{Var } [f_{\hat{\theta}}] (x) := E_{\mathbf{d}} \left[(f(x) - \overline{f(x)})^2 \right]$$

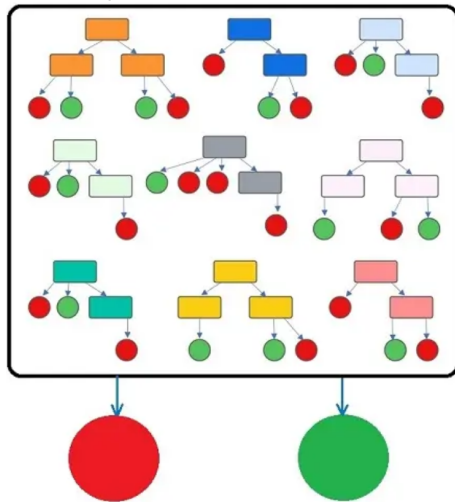


Examples

deep trees



random/boosted forests



This results in an **explainability-fittingness tradeoff!**

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Summary

- ▶ Simply read off
 - ▶ from **case-based reasoning**: training data relevance
 - ▶ from **linear models**: feature relevance
 - ▶ from **decision trees**: reasoning
- ▶ Simulatability-**efficiency** and/or simulatability-**fitness** tradeoffs!

Further Reading

- ▶ Christoph Molnar, *Interpretable Machine Learning* ^[1], Chaps. 6 (linear regression), 7 (logistic regression), 10 (decision trees)
- ▶ Machine Learning Cheat Sheet [↗] for Lecture CS229 – Machine Learning, Stanford University (metrics)
- ▶ Slides on *An Introduction to Ensemble Methods: Bagging, Boosting, Random Forests and More* [↗] by Yisong Yue [↗] (bias-variance trade-off, ensembling, boosting)
- ▶ R. Rojas, A tutorial Introduction to Adaptive Boosting [↗] (FU Berlin, winter term 2009) (boosting)

References

- [1] Christoph Molnar. *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable*. 3rd ed. 2025.
- [2] G. V. Kass. "An Exploratory Technique for Investigating Large Quantities of Categorical Data". In: *Journal of the Royal Statistical Society. Series C (Applied Statistics)* 29.2 (1980), pp. 119–127. ISSN: 0035-9254. DOI: 10.2307/2986296 [↗]. (Visited on 06/15/2024).
- [3] Breiman L. *Classification and Regression Trees*. The Wadsworth & Brooks/Cole, 1984. ISBN: 978-0-412-04841-8. (Visited on 06/15/2024).
- [4] J. Ross Quinlan. *C4.5: Programs for Machine Learning*. Morgan Kaufmann Series in Machine Learning. Morgan Kaufmann, 1993. ISBN: 1-55860-238-0.
- [5] J.R. Quinlan. "Induction of Decision Trees". In: *Machine Learning* 1.1 (Mar. 1986), pp. 81–106. ISSN: 1573-0565. DOI: 10.1023/A:1022643204877 [↗]. (Visited on 06/14/2024).